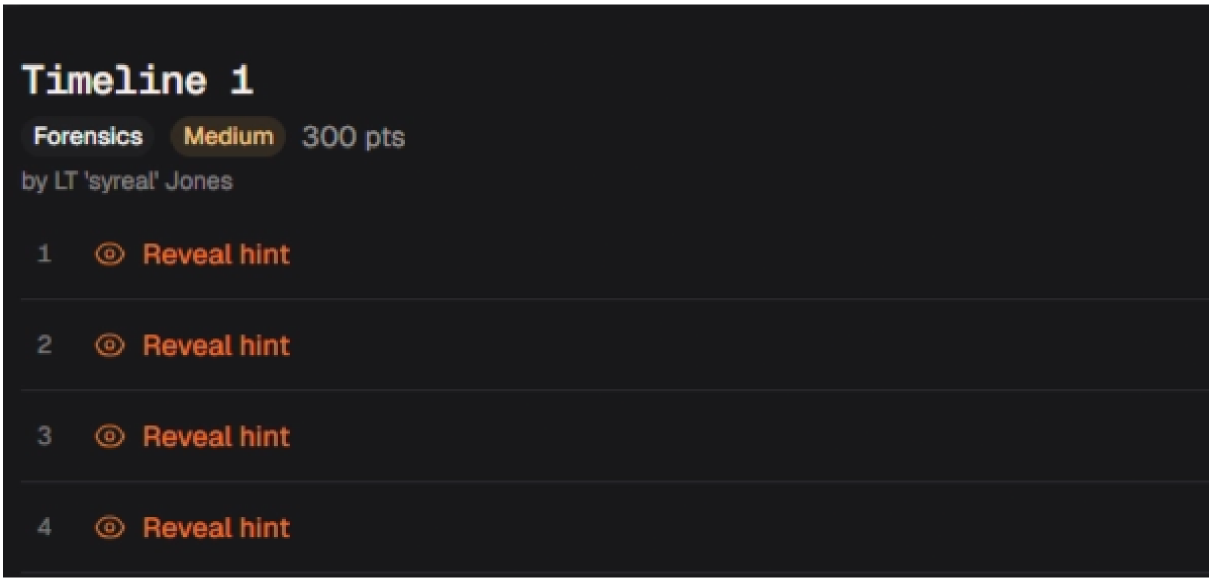
Timeline 1 | picoCTF Writeup

written by Alexander Sapo



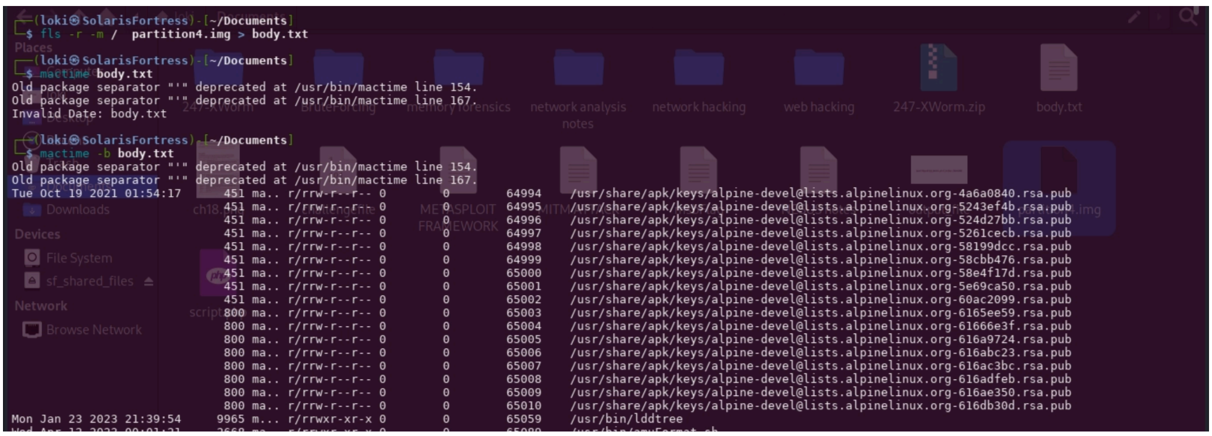
Process:

Since Autopsy felt slow and GUI-heavy for quick triage, I shifted to the Sleuth Kit CLI, which turned out to be significantly faster for targeted analysis.

Tools used

- `fls` — to enumerate and navigate file system structures
- `icat` — to extract file contents by inode
- `mactime` — to build and analyze timeline data

Approach



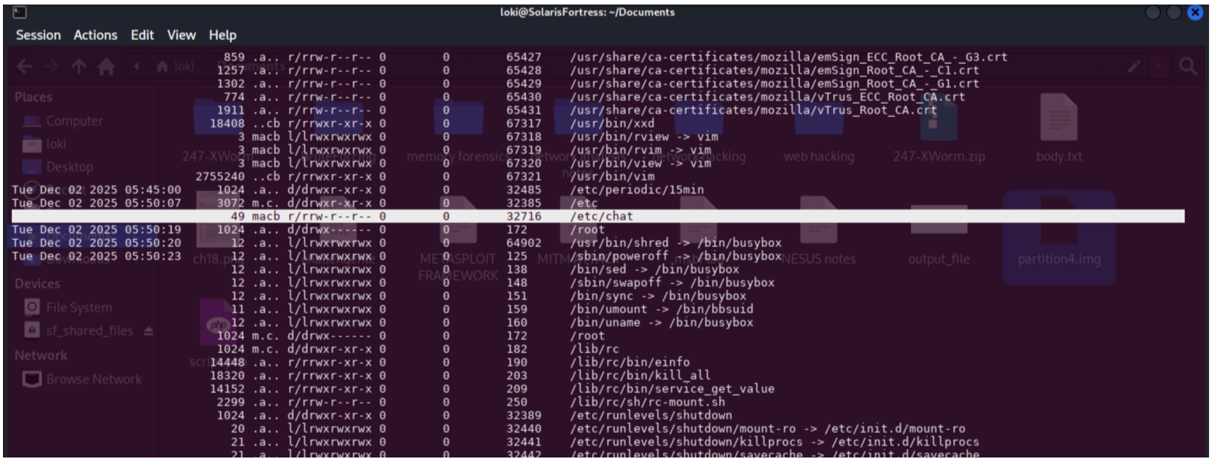
The challenge hinted at examining the timeline, so I generated a body file using `fls`:

```
fls -r -m / partition4.img > body.txt
```

This recursively walks the filesystem and outputs metadata in a format suitable for timeline analysis. I then processed it using `mactime`:

```
mactime -b body.txt
```

From the timeline output, a suspicious entry stood out:




```
49 macb r/rrw-r--r-- 0 0 32716 /etc/chat
```

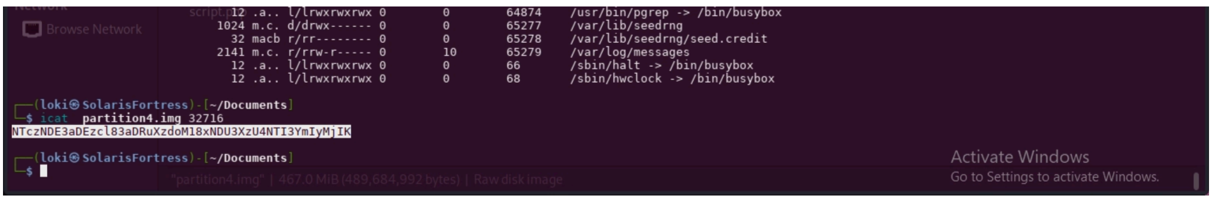
This pointed to an interesting file located at inode `32716`.

File extraction

Using `icat`, I extracted the file directly from the disk image:

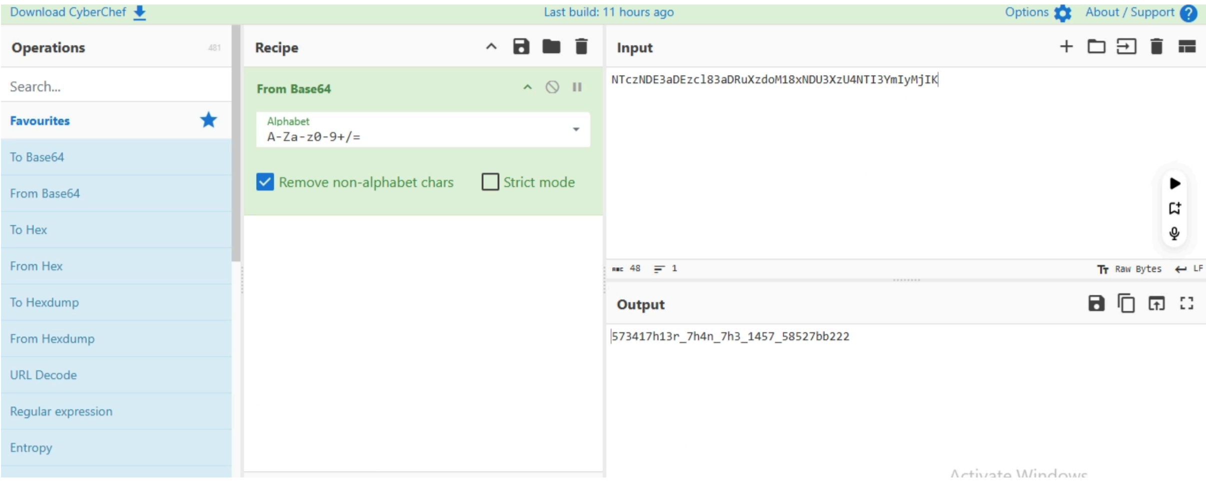
```
icat partition4.img 32716
```

This returned a Base64-encoded string:




```
NTczNDE3aDEzcl83aDRuXzdoM18xNDU3XzU4NTI3YmIyMjIK
```

Decoding



After decoding from Base64, the hidden content was revealed:

```
573417h13r_7h4n_7h3_1457_58527bb222
```

Flag

```
picoCTF{573417h13r_7h4n_7h3_1457_58527bb222}
```